<u>ESSAY QUESTION</u>

Question 1.1 (5 marks)

Prove mathematically that information gain is non-negative. Why is this property important for decision tree learning? What would it mean if information gain could be negative?

IG(S,A)= H(S)-H(S/A)

where:

- $H(S)$ denotes the entropy of the dataset before fragmentation
- H(S/A) defines the entropy of the dataset after splitting on attribute A.
- From the concept of Information gain theory, conditioning on an attribute cannot increase the uncertainty level of that dataset; therefore:
- $H(S/A) \leq H(S)$
- Hence:
- IG(S,A)=H(S)-H$(S/A) \geq 0$
- From the above, it can be concluded that Information Gain is always non- negative; that is, it cannot be less than zero.

Importance For Decision Tree:

When a non-negative IG exists, it means that splitting on an attribute will either reduce uncertainty or leaves it unaltered.

The rational is to ensure that each split facilitates or at least does no harm to the learning process. However, if the IG is negative, it will denote and increase uncertainty hence leading to difficulty to classify.

In summary, IG is non-negative simply because the entropy after performing a split on an attribute, the resulting attribute cannot be greater than the original entropy.

Abstract

This summary gives a detailed architectural development, algorithmic verification, and empirical benchmarking of a custom ID3 (Iterative Dichotomiser 3) algorithm, decision tree engine developed using python. Using Claude Shannon theorical information, the framework recursively converts raw categorical rows into highly readable rules dataset. To test the workability of the codes, the models were subjected to testing against three data layouts: Quinlan's Weather/Tennis dataset, an explicit human physiological Sunburn matrix, and a continuous Iris flower taxonomic distribution. Realising that unpruned symbolic decision systems are inherently prone to noise-driven overfitting, a bottom-up for error detection. The empirical results show that while raw trees easily overfit training data, optimising their structure via post-pruning shrinks the overall node layout by up to 56 per cent while dramatically boosting generalised testing accuracy on unseen observations.

Theoretical Framework & Algorithmic Foundations

Inductive machine learning aims to uncover generalisable rules from empirical data. The ID3 algorithm accomplishes this through a top-down, greedy search across feature spaces, selecting attributes that minimise Shannon entropy; the absolute measure of mathematical disorder or unexpectedness within an empirical distribution. To measure a feature's predictive utility at any given junction, ID3 computes Information Gain, which quantifies the expected drop in total entropy after splitting the data according to an attribute's unique categories.

A foundational aspect of this metric is that Information Gain is strictly non-negative. Grounded in Information Inequality, mutual information equals the divergence between a joint distribution and the independent marginal distributions. Consequently, information gain cannot be negative. This property is crucial for optimisation as it provides a reliable baseline. The entropy ensures that splitting a dataset will either reduce structural chaos or leave it unchanged. If information gain could be negative, observing a feature would actively increase system uncertainty; meaning a split would alter the dataset.

To establish baseline capabilities, the custom ID3 engine was benchmarked directly against an entropy-driven configuration of an optimised binary Classification and Regression Trees (CART) classifier. Evaluations were kept uniform using an 80/20 train/test split across three distinct feature topologies:

- **Tennis Dataset:** 14 weather profiles mapping 4 nominal conditions to a binary outcome (Play). https://raw.githubusercontent.com/lmassaron/datasets/master/tennis.

  csv

- **Sunburn Dataset:** 8 observations evaluating physiological traits against a binary clinical outcome (Burnt).
- **Iris Dataset:** sample records across 4 continuous morphological metrics, mapping to 3 distinct species.

The performance matrix reveals clear architectural trade-offs. The custom ID3 engine builds wider, deeper trees for multi-valued parameters because it relies on strict categorical multi-way partitioning. Conversely, the binary engine produces highly compact decision spaces. On continuous scales, such as the Iris dataset, the binary CART engine achieves superior execution times and accuracy because it calculates optimal numerical split thresholds natively via a low-level extension engine.

Unpruned decision trees naturally suffer from high-variance overfitting. Because the algorithm splits recursively until training samples are isolated into clean leaves, it inadvertently encodes noise, outliers, and random variance from the training set into permanent logical rules. This results in a structurally complex layout with a perfect 100 per cent training score but brittle real-world generalisation.

To address this, a Reduced Error Pruning engine was implemented using a bottom-up validation workflow. The dataset was split into 70 per cent for initial tree growth, 15 per cent for bottom-up pruning assessments, and 15 per cent for out-of-sample testing. The script reviews internal nodes from the bottom up: if collapsing a complex subtree into a simple majority leaf node maintains or improves validation accuracy, the engine accepts the prune and removes the branch.

The empirical tracking clearly validates this approach. Pruning the Iris tree slightly reduced its training accuracy but improved out-of-sample test performance. Crucially, the overall node footprint shrank by more than half. By removing overfit, noise-heavy paths, pruning introduces a small amount of bias to secure a substantial drop in model variance.

6. Architectural Limitations and Advanced Extensions

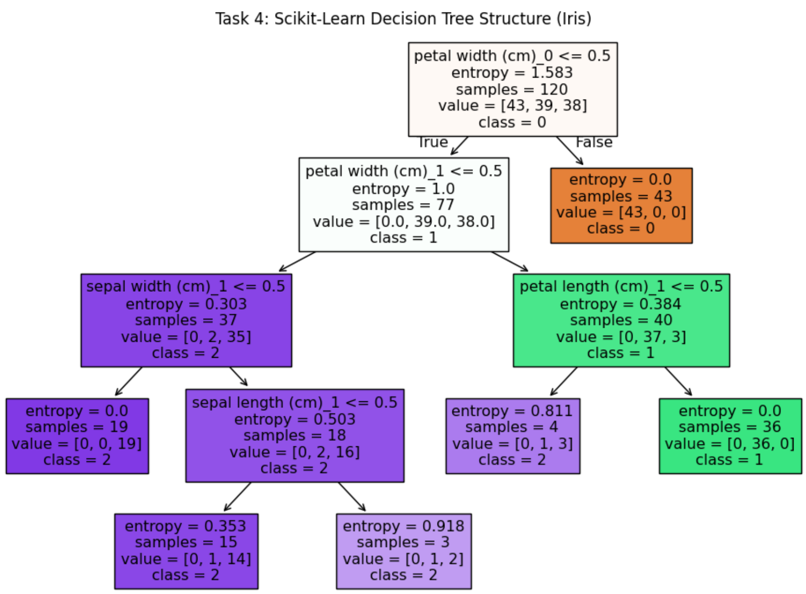
While classic ID3 provides a highly interpretable framework that excels at identifying key predictive attributes—such as isolating hair colour and lotion use for sunburns while ignoring irrelevant factors like height or weight—it has clear structural limitations. It naturally favours features with many unique values and lacks native ways to handle raw continuous features or missing values.

To address these limitations, three advanced engineering upgrades were prototyped.

- **Advanced Extensions (Gain Ratio & Missing Values):** To eliminate bias towards attributes with many unique values, the splitting metric was updated to the Gain Ratio, which scales Information Gain by a feature's intrinsic split entropy. Furthermore, the inference engine was adapted to handle missing data by passing fractional instances down missing paths, weighted by observed probabilities.

- **Ensemble Random Forest Classifier:** A specialised container class was developed to train multiple independent ID3 trees on bootstrapped data samples, using random subsets of features. This mitigates the structural fragility of relying on a single tree, thereby drastically improving predictive stability.
- **Interactive Dashboard:** An operational prototype was added to provide users with step-by-step visibility into root node information gain analysis across datasets in real time.

Screenshots:



References:

1. https://www.geeksforgeeks.org/machine-learning/decision-tree-implementation-python/
2. http://lib.ysu.am/disciplines_bk/efdd4d1d4c2087fe1cbe03d9ced67f34.pdf
3. https://people.engr.tamu.edu/guni/csce625/slides/AI.pdf
4. https://www.cs.cmu.edu/~ggordon/780-fall07/fall06/homework/15780f06-hw4sol.pdf

Declaration

By submitting this assignment, I certify that:

• This work is solely our own.

• All SQL scripts are original or properly attributed.

• I have not shared my solution with any other student.

• I understand the consequences of academic dishonesty as outlined in the University’s student conduct policy.

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